ADVANCED NLP · FINAL EVALUATION

Detecting Cross-Turn Rhetorical Fallacies in Discourse

Motte-and-Bailey · Strawman

Presenter: Aaditya Sharma | random.strings

The key signal lives between claims

Sentence-level classification is not enough.

Each claim can sound reasonable on its own. The fallacy is in how a later claim retreats from, or misrepresents, an earlier one.

Motte-and-bailey: claim strongly, retreat to a safer claim when challenged, act as if nothing changed.

Strawman: restate the other person's claim as something stronger, then attack that.

A REAL EXAMPLE FROM r/CHANGEMYVIEW

Claim A

If these measures are expected and treated as reasonable, it serves to make the crime more acceptable…

Challenge

…this is hogwash. Leaving my doors and windows unlocked is not going to reduce theft.

Claim B

I never said that leaving them unlocked reduces theft… Essentially, that locking doors increases crime.

Neither A nor B is fallacious alone. The retreat is only visible in the pair.

Related work: fallacy labels vs. dialogue relations

What exists, and what remains unexplored

Fallacy detection

Labels one comment or text span. Strawman already exists as a class, but the claim being misrepresented is not part of the unit.

MAFALDA · LOGIC · CoCoLoFa · Ramponi et al. · Sahai et al.

Dialogue relations

Annotate inference, conflict and rephrase between claims across turns, but not whether the relation is fallacious.

QT30 / DialAM-2024 · Koszowy et al. · Uberna et al.

Theory → gap

Strawman as misuse of rephrase (Visser et al., 2018): theory only, no dataset or classifier. No computational motte-and-bailey work found.

Visser et al. · Shackel (2005)

Our angle: an annotated, evaluated set of cross-turn claim pairs for both fallacies, with models tested on it.

Shortcut risk is known: NLI models exploit annotation artifacts (Gururangan et al., 2018; McCoy et al., 2019); fallacy detectors flag valid arguments that share a fallacy's scheme (Singh et al., 2026).

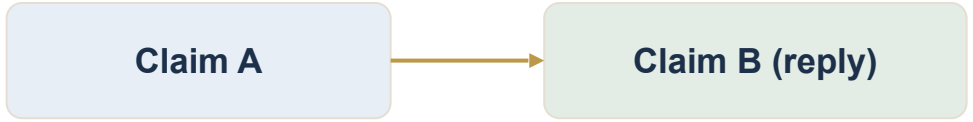
Two fallacies, two relational structures

01 MOTTE-AND-BAILEY



Same speaker writes A and B; C comes from someone else. Is B a weaker claim offered as if it still defends A?

02 STRAWMAN



B is a direct reply from another speaker. Does B distort A into something stronger or different, and argue against that?

Two separate binary tasks (MOTTE_BAILEY vs NONE, STRAWMAN vs NONE). Speaker identity and the challenge are fixed by the shape, so a 3-way classifier could tell the shapes apart without reading the claims.

Central question: relation or surface cue?

Do models detect these fallacies by relating the two claims, or by surface cues such as the phrases used to find the candidates?

Compare

Feature model, fine-tuned cross-encoders and zero-shot LLMs on identical folds

Ablate inputs

Reply B alone vs. the full exchange; delete trigger phrases at test time; minimal-edit flip pairs

Look inside

Layer-wise probing of frozen encoders, NLI-trained vs. plain

Test the theory

Extract the attributed / denied claim, then check it against A with NLI

Why it matters: every real candidate was selected because its reply contains a phrase like “I never said” or “so you're saying”.

From a discussion corpus to labeled claim pairs

r/ChangeMyView (Winning Arguments corpus, via ConvoKit)

293,297
utterances in 3,051 conversations

→

376,714
structural candidates (reply tree)

→

5,078
contain a trigger phrase

→

400
sampled for claim extraction

→

317
usable condensed claims

→

188
hand-labeled real pairs

TRIGGER-PHRASE FILTER ON B

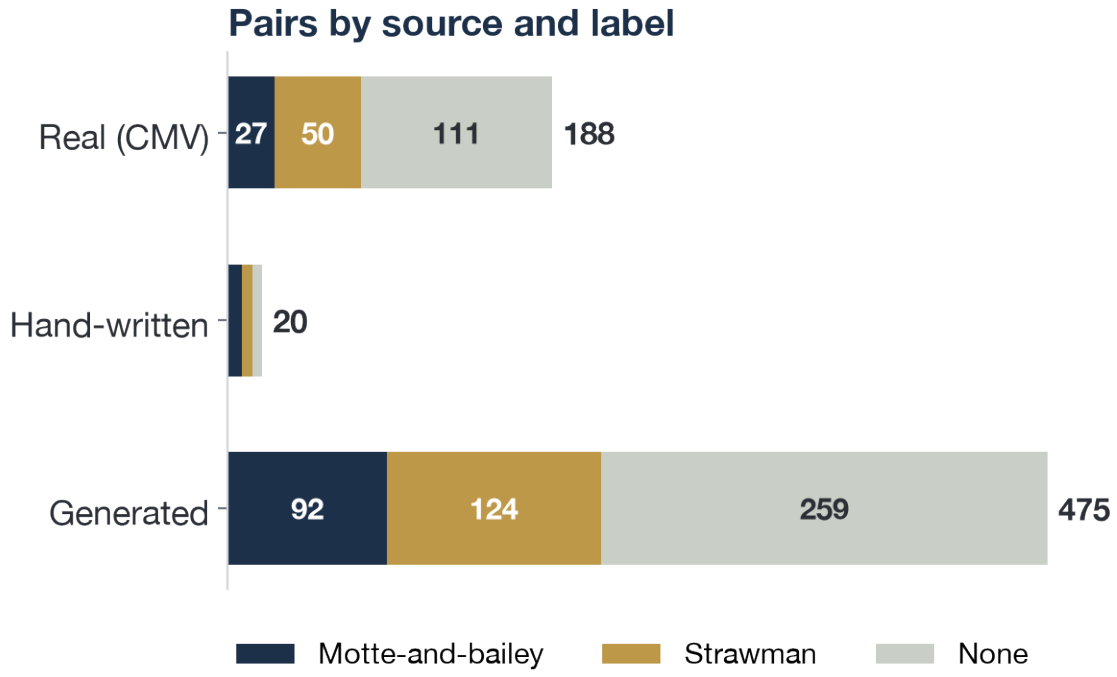
- 16 denial phrases for motte-and-bailey: “I never said”, “what I meant”, “all I'm saying is” …
- 12 attribution phrases for strawman: “so you're saying”, “are you saying” …
- Long comments condensed to short claims A, C, B by an LLM

ANNOTATION

- One annotator; labels MOTTE_BAILEY, STRAWMAN, NONE, UNCERTAIN
- 20% of rows UNCERTAIN (claims alone not enough) and excluded
- 188 pairs from 169 conversations

Selection bias: the phrase is in 96% of motte-and-bailey and 89% of NONE (same shape) real rows. It can't separate the classes, but a model can still lean on it.

683 claim pairs: real, hand-written and generated



GENERATED VARIANTS (CLAUDE SONNET 5.5)

227 **Paraphrase**
same exchange reworded, same label

228 **Topic transfer**
same logical move on a different everyday topic

20 **Flip**
A and C copied, B minimally edited so the label changes

Real seeds get a paraphrase and a topic transfer that keep the human label; flips only come from hand-written seeds.

Two tasks: motte-and-bailey 353 pairs (36% positive), strawman 330 pairs (55% positive). Every variant stays in the same CV fold as its source.

Four model families, one protocol

Feature model

Frozen NLI (DeBERTa-v3-large) gives entail / neutral / contradict for A→B and B→A, plus 5 lexical features

→ **XGBoost**

Fine-tuned cross-encoder

DeBERTa-v3-large on [A] [SEP] [C] [SEP] [B]; NLI-initialised vs. plain; B-only ablation

→ **6 epochs, 3–5 seeds**

Zero-shot LLMs

Qwen2.5-7B and 32B on all 683 pairs; Claude Opus 5.5 on 111 originals. Definitions only, no examples

→ **label only (no AUC)**

Two-stage pipeline

LLM extracts the claim B attributes to (or denies from) A; frozen NLI checks P(A ⇒ X)

→ **no training, threshold 0.5**

Why NLI? A strawman attributes something A doesn't support; a retreat changes the scope of A. Both look like inference relations between claims.

Evaluation design and what the scores mean

5-fold
CV grouped by conversation

97 / 101
groups (MB / SM)

3–5
seeds per configuration

2,000
cluster-bootstrap resamples

GUARDRAILS

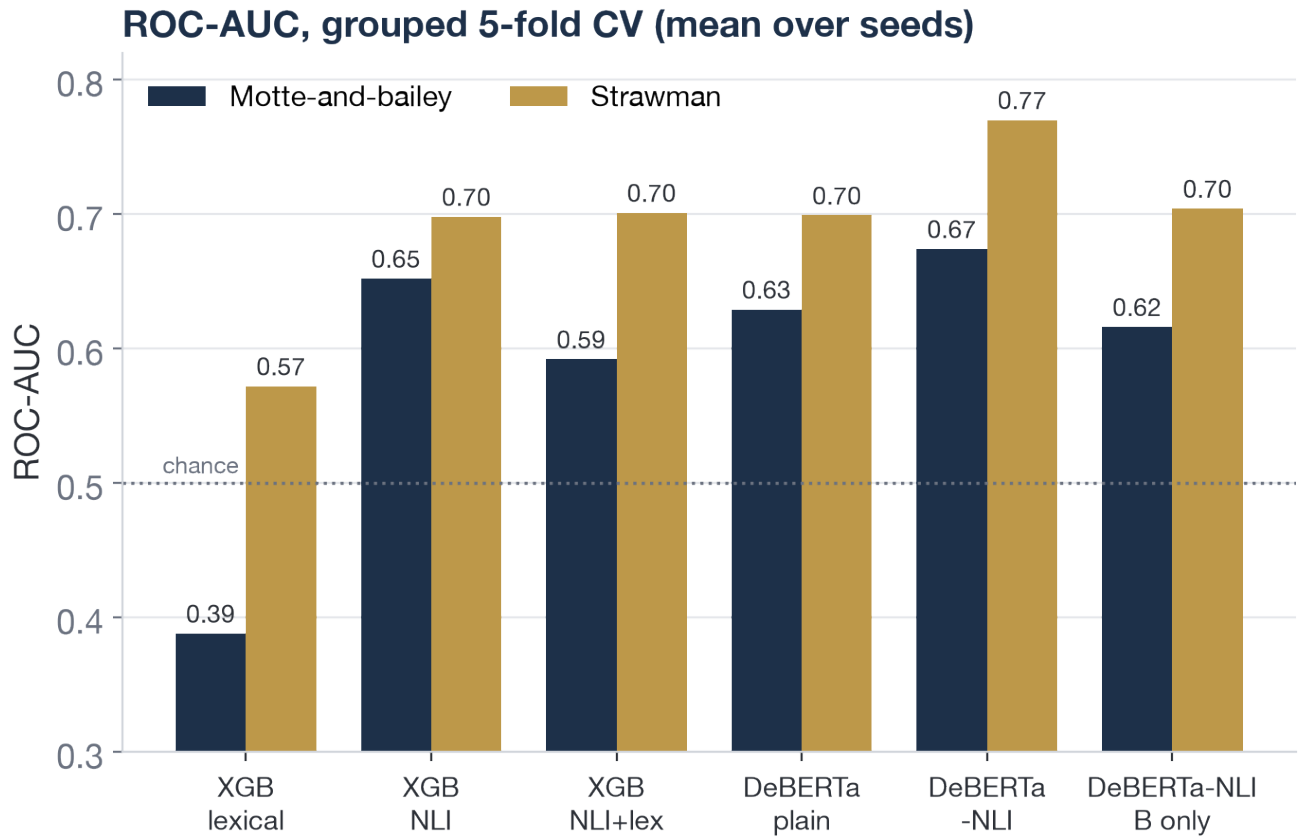
- An example and its variants always share a fold: no model sees a paraphrase of its test pair
- All trained models use identical folds, so comparisons are paired
- A difference counts only if the 95% interval excludes 0

READING THE NUMBERS

	Majority acc.	Always-fallacy F1
Motte-and-bailey	0.640	0.529
Strawman	0.545	0.706

ROC-AUC is the main metric for motte-and-bailey: accuracy and F1 there barely move above these baselines.

Fine-tuning ranks best, but isn't more accurate



0.770 / 0.674

best AUC, strawman / motte-and-bailey (fine-tuned DeBERTa-NLI)

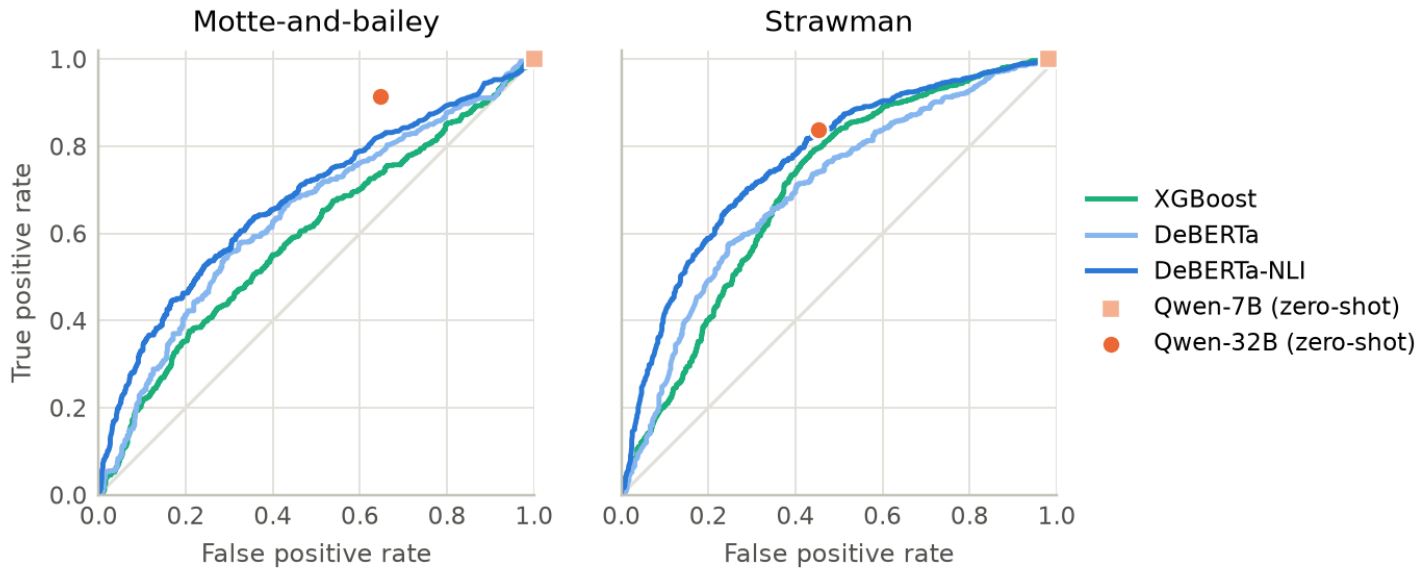
+0.009 / +0.007

accuracy gain over XGBoost; both 95% CIs include 0

0.388

lexical-only AUC on motte-and-bailey: worse than chance

A strong zero-shot LLM is a serious baseline



ROC curves of the trained models; zero-shot LLMs are single points (labels only).

Caveat: the Claude run is a single run on a small subset, and the same model family extracted the claims and generated the variants.

Qwen2.5-7B

Labels every pair a fallacy: exactly the always-fallacy baseline.

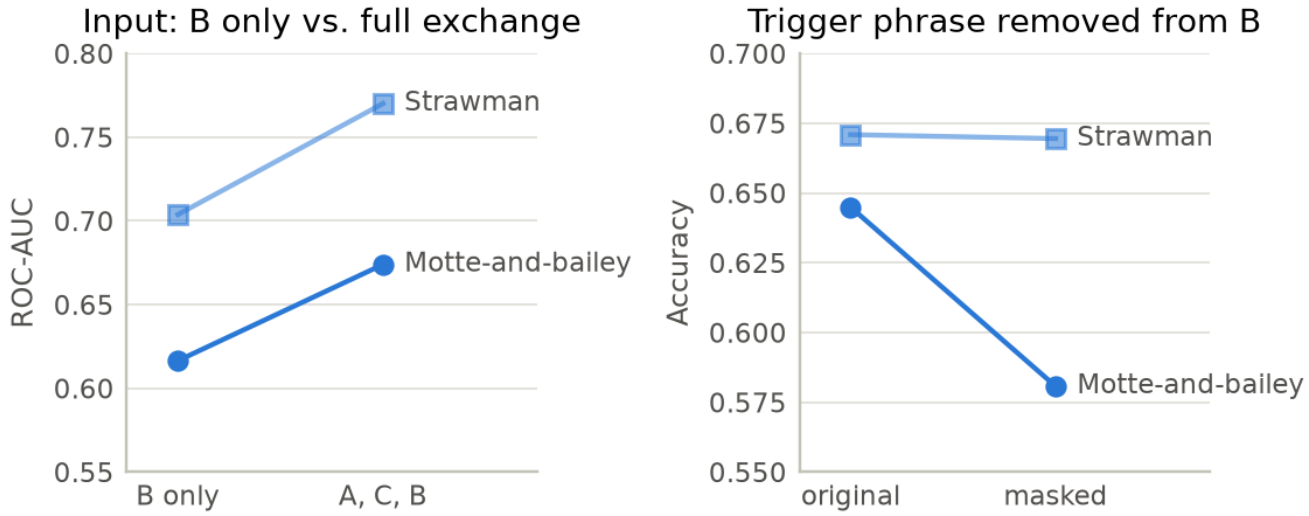
Qwen2.5-32B

Matches fine-tuning on strawman (0.706 acc., Δ −0.009, n.s.) but flags 65% of motte-and-bailey negatives.

Claude Opus 5.5

Best on the 111 originals: 0.638 / 0.736 accuracy vs. 0.576 / 0.592 for fine-tuned DeBERTa-NLI.

Context helps; motte-and-bailey leans on the trigger phrase



Fine-tuned DeBERTa-NLI. Right panel: accuracy on rows whose B contains a trigger phrase.

+0.063

accuracy from the full exchange vs. B alone (MB); the only significant difference

0.645 → 0.581

MB accuracy when the phrase is deleted; strawman unchanged (0.671 → 0.670)

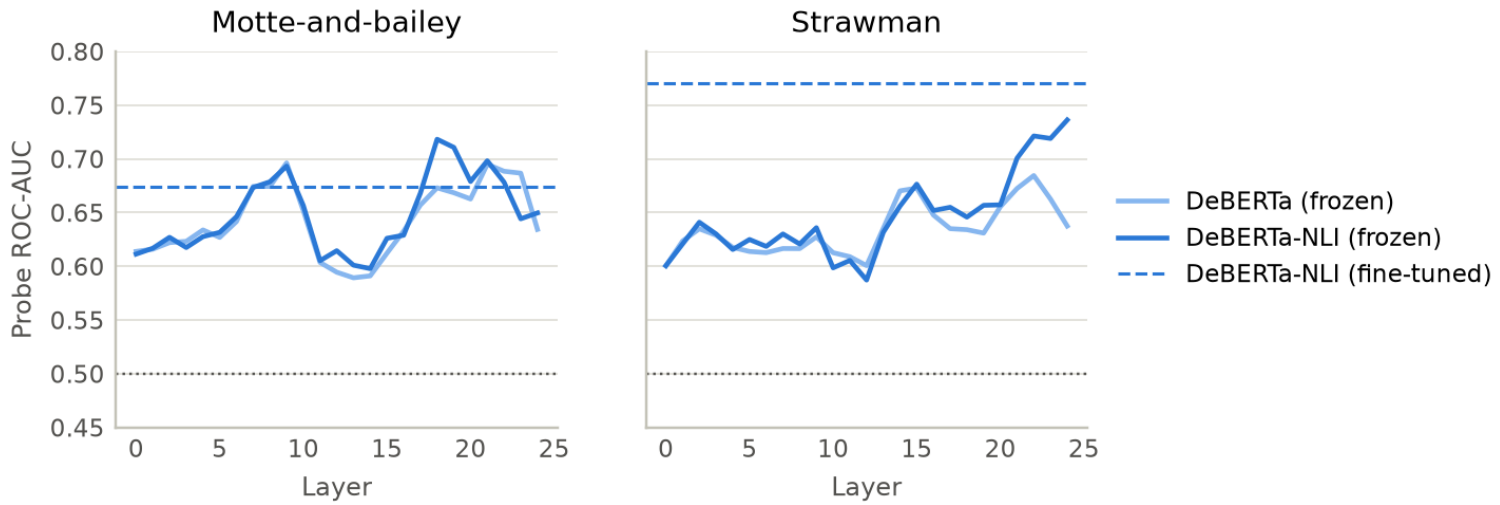
Plain DeBERTa does not show the drop (0.625 → 0.641), so the reliance is clear for the best model, not universal.

m026 · GOLD: NONE

B: "I'm fully aware of how the word is used. All I'm saying is that usage is often incorrect."

B restates A rather than retreating. Fine-tuned model says MB in 2/3 seeds; with "All I'm saying is" deleted it says NONE in every seed.

Strawman signal sits in the NLI encoder's last layers



Logistic-regression probe per layer of frozen encoders; dashed = fine-tuned DeBERTa-NLI.

≈ 0.60

AUC from the embedding layer alone: word content already separates classes a bit

0.736 vs 0.638

strawman, last layer: NLI encoder vs. plain encoder

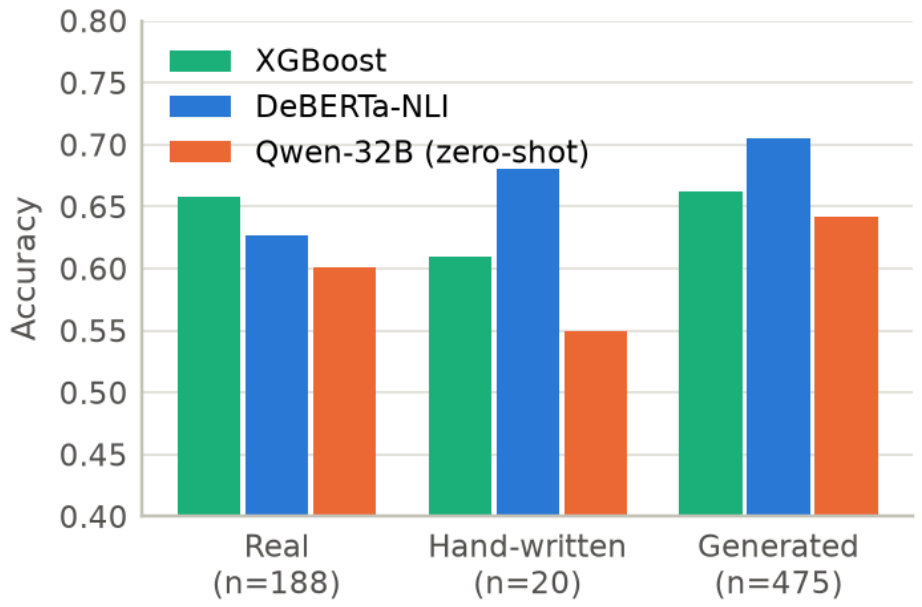
0.718 ≥ 0.674

motte-and-bailey: frozen NLI layer 18 matches full fine-tuning

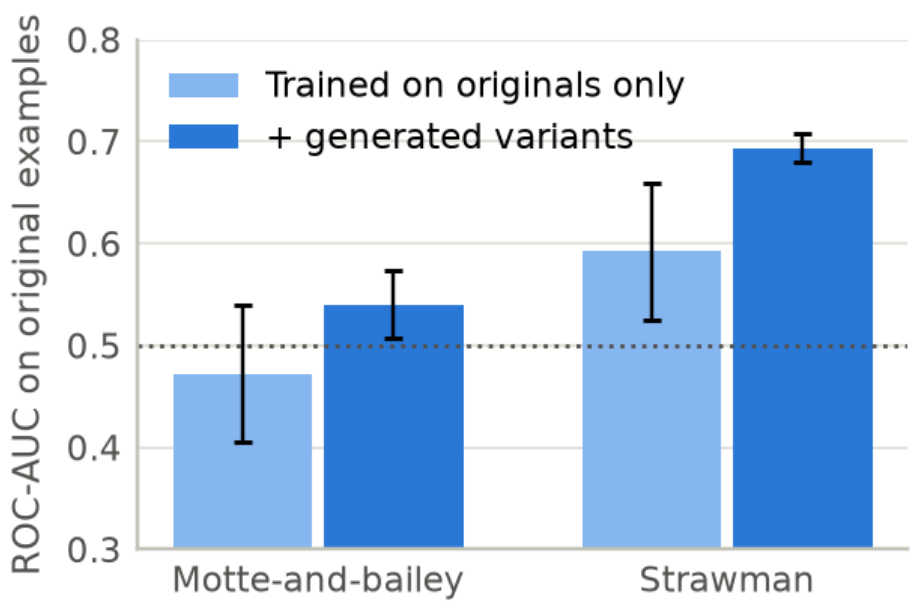
NLI pretraining helps strawman only: fine-tuned AUC 0.770 vs. 0.699 from plain DeBERTa (accuracy +0.045, borderline). For motte-and-bailey the two initialisations are the same (+0.005).

On real exchanges, motte-and-bailey is close to chance

ACCURACY BY DATA SOURCE



AUC ON THE 208 ORIGINAL PAIRS



Real CMV pairs: fine-tuned 0.627 vs. XGBoost 0.657. The advantage comes from generated pairs.

Variants help strawman on originals (0.592 → 0.694); motte-and-bailey stays at 0.47–0.54.

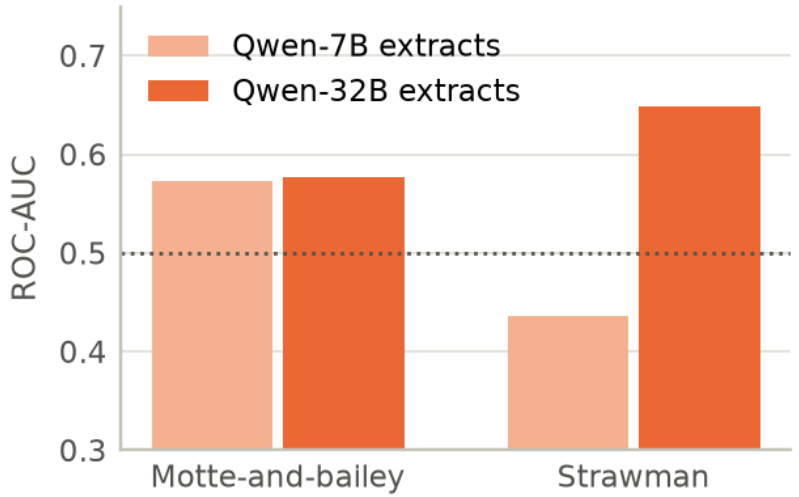
Extract, then verify: works for strawman with a strong extractor

THEORY-DRIVEN

STRAWMAN AS MISUSE OF REPHRASE

LLM reads A, B → **Writes claim X that B attributes to A** → **NLI: P(A ⇒ X)**

STRAWMAN if P(A ⇒ X) < 0.5. No training, threshold fixed in advance.



s001 · GOLD: NONE

B: “So you're saying that just by being randomly born in a country, I owe it something?”

Pointed but fair. Every trained model and Qwen2.5-32B say STRAWMAN; the pipeline finds P(A ⇒ X) = 0.67 and correctly says NONE.

Motte-and-bailey rules fail (AUC 0.49–0.58): stage 1 finds a denial in almost every pair, as the selection guarantees.

Two fallacies, two different answers

Diagnostic	Strawman	Motte-and-bailey
Full exchange vs. B only	+0.066 AUC	+0.063 accuracy (significant)
Trigger phrase deleted	no change (0.671 → 0.670)	drops (0.645 → 0.581)
NLI pretraining	helps (AUC 0.770 vs 0.699)	no difference
Frozen-encoder probing	peaks in last NLI layer	frozen layer 18 = fine-tuned
Original examples only	AUC 0.694 with variants	near chance (0.47–0.54)
Extract-then-verify	AUC 0.648 (32B extractor)	at or near chance

Strawman: a relational, inference-related signal that does not depend on the attribution phrases.

Motte-and-bailey: some use of context, but partly a trigger-phrase shortcut, and not learned on real data.

Limitations and next steps

LIMITATIONS

- Only 188 real labeled pairs; learning curves still rising
- Single annotator, no agreement measure; labels on LLM-condensed claims
- Every real candidate has a trigger phrase: fallacies without one are missing
- 475 generated pairs not checked by hand; models score higher on them
- One forum (r/ChangeMyView), moderated and good-faith

NEXT STEPS

- Label the remaining 81 extracted pairs and add a second annotator
- Mine candidates without trigger phrases, e.g. replies that call out a strawman
- Hand-written contrast sets: fair replies with a trigger, distortions without one
- Combine the two-stage signal with learned models; better claim extraction
- Out-of-domain test on debate data such as QT30

CONCLUSION

What we learned

01 FORMULATION

Cross-turn claim pairs make both fallacies annotatable and learnable above chance: 683 pairs, best AUC 0.770 / 0.674.

02 EVIDENCE

Strawman detection uses an inference-related relation between claims. Motte-and-bailey partly relies on denial phrases and is near chance on real pairs.

03 LESSON

Pooled scores hide shortcuts. Evaluate on real examples, split by source, with negatives that share the surface cue.

Thank you · Questions?

github.com/Ad-S/relational-fallacy-detection

Backup: full results table

Grouped 5-fold CV on all 683 pairs, mean over seeds; F1 for the fallacy class

Model	MB acc.	MB F1	MB AUC	SM acc.	SM F1	SM AUC
Majority class	0.640	0.000	0.500	0.545	0.706	0.500
XGBoost, lexical only	0.550	0.135	0.388	0.561	0.626	0.572
XGBoost, NLI only	0.666	0.468	0.652	0.673	0.717	0.698
XGBoost, NLI + lexical	0.638	0.412	0.592	0.682	0.725	0.701
DeBERTa plain, fine-tuned	0.646	0.531	0.629	0.652	0.692	0.699
DeBERTa-NLI, fine-tuned	0.665	0.538	**0.674**	0.701	0.736	**0.770**
DeBERTa-NLI, B only	0.587	0.514	0.616	0.655	0.699	0.704
Qwen2.5-7B, zero-shot	0.360	0.529	–	0.555	0.710	–
Qwen2.5-32B, zero-shot	0.555	0.596	–	0.706	0.757	–

Paired accuracy differences (95% cluster-bootstrap CI): DeBERTa-NLI − XGBoost +0.009 [−0.063, +0.083] MB, +0.007 [−0.062, +0.077] SM; full − B only +0.063 [+0.007, +0.116] MB.